What Stuff Do I Need?

https://education.ti.com/en/product-resources/download: instructions for downloading stuff onto your calculator. Download TUNING, CUSTOM, and INSTR4MT onto your calculator

You'll need a 2.5 mm male to 3.5 mm female audio jack adapter. You'll use this to plug your headphones (or other audio device) into your calculator.

A TI-84+ or TI-83+ graphing calculator.

Running the Program

You should have downloaded the programs TUNING, CUSTOM, and INSTR4MT onto your calculator.

TUNING and CUSTOM are TI-BASIC programs. Run TUNING, then CUSTOM, before you run INSTR4MT. You'll only need to run these two again if you change your L1, L2, or E variables. TUNING should take around 10-15 seconds to complete.

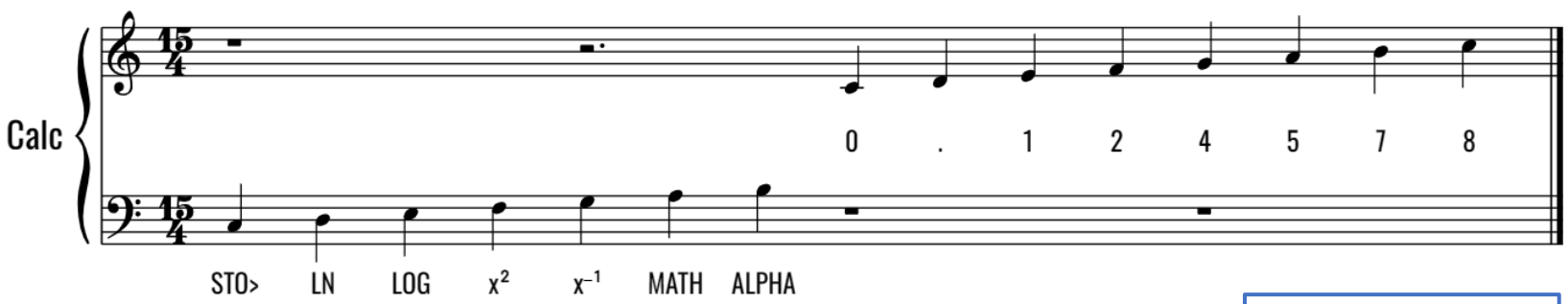
INSTR4MT is an assembly program. Run this, and your calculator is an instrument! If when you run it, you see ERR: INVALID DIM, try running TUNING and CUSTOM again. Put in the audio adapter <u>AFTER</u> starting the program.

Look up how to run assembly and TI-BASIC programs if you need to.

How to use *INSTR4MT*

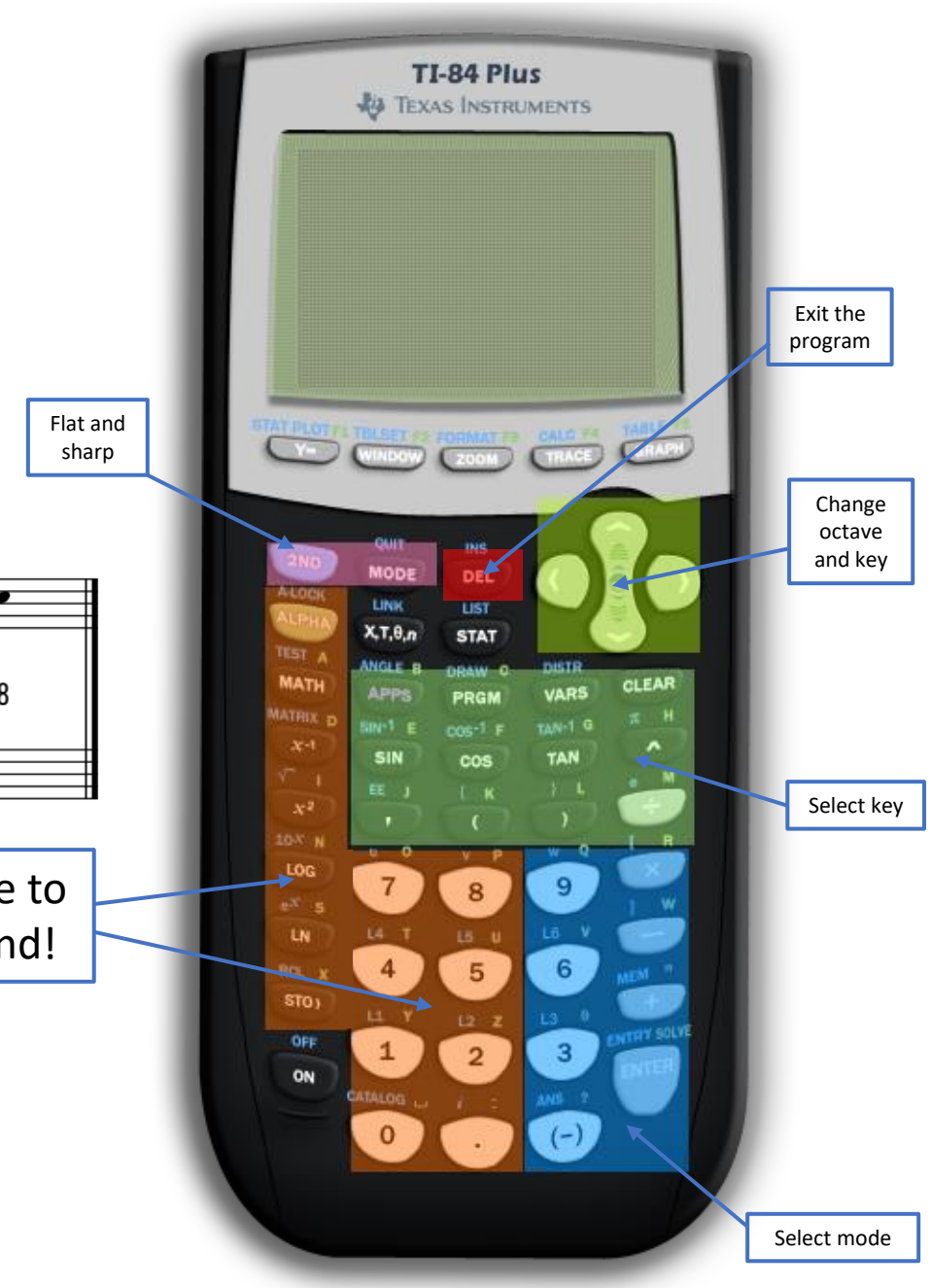
There are 88 keys on a piano, and only about 54 keys on my TI-84+. So, instead of mapping one note to each button chromatically, the instrument works by having 15 playable buttons and the ability to switch between musical modes, keys, and octaves.

Choose the musical mode with the blue buttons, the key with the green buttons, and the octave with the up and down buttons. Play the notes of the scale you've chosen using the orange buttons.



[(-)] Ionian [ENTER] Dorian [3] Phrygian
[+] Lydian [6] Mixolydian [-] Aeolian
[9] Harmonic Minor [x] Custom Mode

[APPS] F	[PRGM] F#/Gb	[VARS] G	[CLEAR] G#/Ab
[SIN] C#/Db	[COS] D	[TAN] D#/Eb	[^] E
[,] A	[(] A#/Bb	[)] B	[÷] C



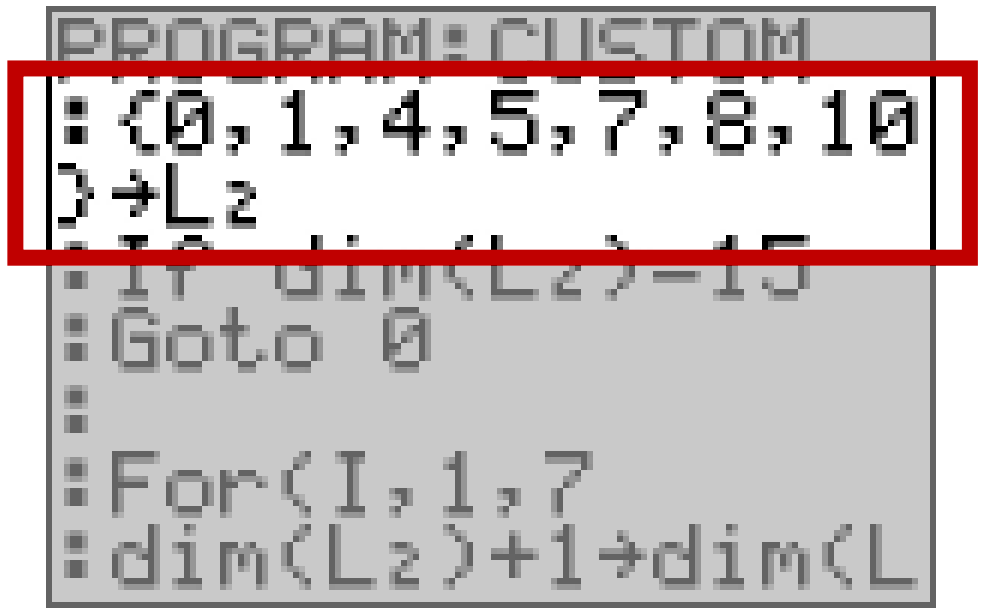
Press [2ND] to bring the note currently being played down a half step. Press [MODE] to bring the note currently being played up a half step.

Use the up and down buttons to scroll through octaves and use the left and right buttons to move the key down or up by a half-step.

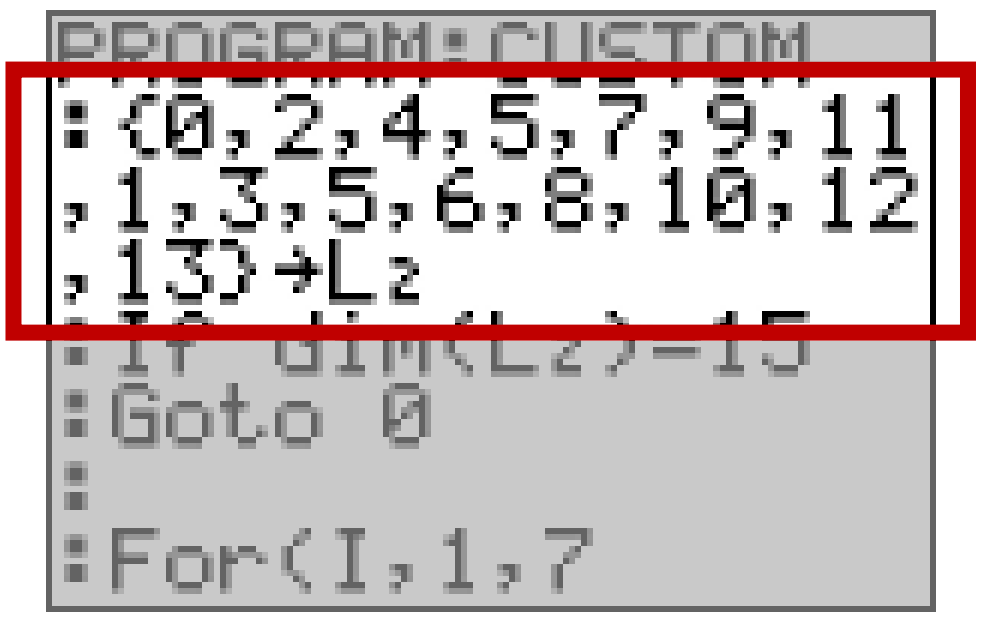
On start-up pressing [8] will play an A of approx. 440 Hz
Press [DEL] to exit the program

Notes
- A 2.5mm to 3.5mm headphone jack Is required to use this program.
- Type "Asm(prgmINSTR3MT)" on your Device to start the program.
- Start the program before putting in your headphones
- Only confirmed to work on my TI-84+

How to add a custom mode/scale



Press the PRGM button on your calculator and edit the program called CUSTOM. The mode is stored in L2, and each value in L2 is how many semitones away from the root a note should be. For example, on the image on the top left, this mode's 5th note is 7 semitones from the root. All the rest of the program does is add 8 more notes to L2 an octave above the first 7 notes put into it at the beginning of the program.



If you put 15 notes in L2, then the first 7 will be used by the low note buttons (STO> to ALPHA) and the last 8 will be used by the high note buttons (0 to 8). You can do some wacky stuff with this. If you set L2 like the bottom left image, your low note buttons will play a major scale and your high note buttons will play a major scale in the same octave but one semitone higher. Neat!

You could make the low buttons play the high notes and the high buttons play the low notes if you want. Get creative!

Tuning your calculator and microtonality

```
PROGRAM:TUNING
:1.003→C
:27.50→A
:12→E
:0→I
:0→J
:1→X
:6615000*C→T
```

BONUS COOL STUFF FOR TI-BASIC PROGRAMMERS!!!

TUNING can only generate equal temperament stuff, but if you can program in TI-BASIC, there's nothing stopping you from making your own tuning program that does whatever you want. Just fill up L1 and set E to however many steps you want in your twisted version of an octave, and boom! New tuning! If you want a note in L1 to have a frequency of *f,* put the result of this formula in that spot of L1:

$$\frac{\frac{6615000 \cdot C}{2 \cdot f} - 795}{26}$$

To tune your calculator, all you need to do is mess around with the C variable in the program called TUNING. Make it larger to make the instrument flatter and make it smaller to make the instrument sharper. Run TUNING each time you change it. I have my C set to 1.003, but each calculator will probably need a different value.

Guess what. MICROTONALITY!!!! When E is set to 12, like it is by default, the octave will be divided into 12 equal parts. But you can change this to any number you like. You could divide it into 19 parts, or 24 parts, or whatever you want. I suggest making a custom mode that will work well with the new tuning. Run TUNING each time you change E. It will take longer for higher values of E.

Set E to 31 in TUNING and set L2 to {0,5,9,14,18,23,27} in CUSTOM, and you can use the custom mode (accessed by pressing [x]) to play the Heptatonic Mohajira scale the YouTuber Levi McClain talks about in his microtonal Hey Ya video! First run TUNING, then run CUSTOM.

Also, when using high values of E, the step size is smaller. If you hold a note spam one of the step up/down buttons ([2nd] or [MODE]) *really fast,* it kinda sounds like vibrato.

Wait, what order do I run stuff in? I’m confused.

TUNING sets two values called E and L1. CUSTOM sets a value called L2, and needs E to set L2 properly. INSTR4MT needs E, L1, and L2 to be set properly to work.

So, if you haven’t done anything that messes with L1 or E, you don’t need to run TUNING again. You should run CUSTOM again if you change L2 or E.